

Life Processes in living organisms Part -2

I] Textbook Questions:

★ 1 Mark Objectives: ★

1. Give the names.

- a. Hormones related with male reproductive system. **Ans:** Testosterone, FSH and LH
- b. Hormones secreted by ovary of female reproductive system. **Ans:** Estrogen and Progesterone
- c. Types of twins. **Ans:** Monozygotic and Dizygotic
- d. Any two sexual diseases. **Ans:** Syphilis, gonorrhea [2025]
- e. Methods of family planning. **Ans:** Vasectomy, Tubectomy, condom, contraceptive pills.

2. Fill in the blanks.

- a. In humans, sperm production occurs in the organ **testes**. [2020, 2025]
- b. In humans, **Y** chromosome is responsible for maleness.
- c. In male and female reproductive system of human, **bulbourethral gland (male) and Bartholin's gland (female)** is same.
- d. Implantation of embryo occurs in **uterus**.
- e. **Asexual** type of reproduction occurs without fusion of gametes.
- f. Body breaks up into several fragments and each fragment starts to live as a new individual. This is **Fragmentation** type of reproduction.
- g. Pollen grains are formed by **meiotic** division in locules of anthers.

★ 2 Marks Questions: ★

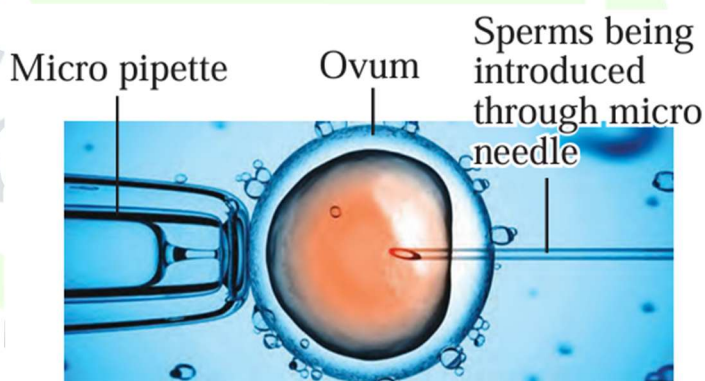
1. Answer the following questions in short.

- a. Explain the concept of IVF. [2020]

Ans:

In Vitro Fertilization (IVF):

- In this technique, fertilization is brought about in the test-tube and the embryo formed is implanted in uterus of woman at appropriate time.
- IVF technique is used for having the child in case of those childless couples who have problems like less sperm count, obstacles in oviduct, etc.



3.22 Fertilization in a test tube

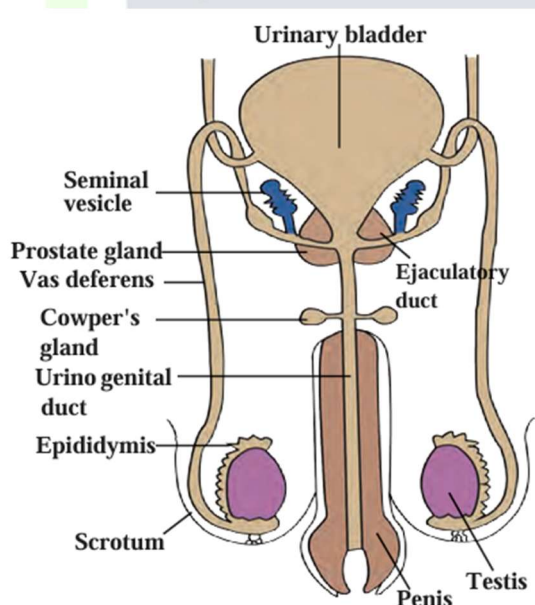
2. Modern techniques like surrogate mother, sperm bank and IVF technique will help the human beings. Justify this statement.

Ans:

1. Many couples cannot have children due to various reasons.
2. In case of women, irregularity in menstrual cycle, difficulties in oocyte production, obstacles in the oviduct, difficulties in implantation in uterus and many other reasons are responsible for this.
3. Absence of sperms in the semen, slow movement of sperms, anomalies in the sperms are the reasons in case of males.
4. But now with the help of advanced medical techniques like IVF, Surrogacy, Sperm bank the childless couples can have a child.

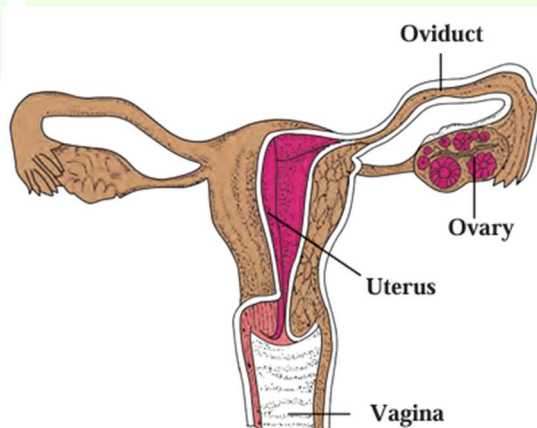
2. Sketch the labelled diagrams.

a. Human male reproductive system.



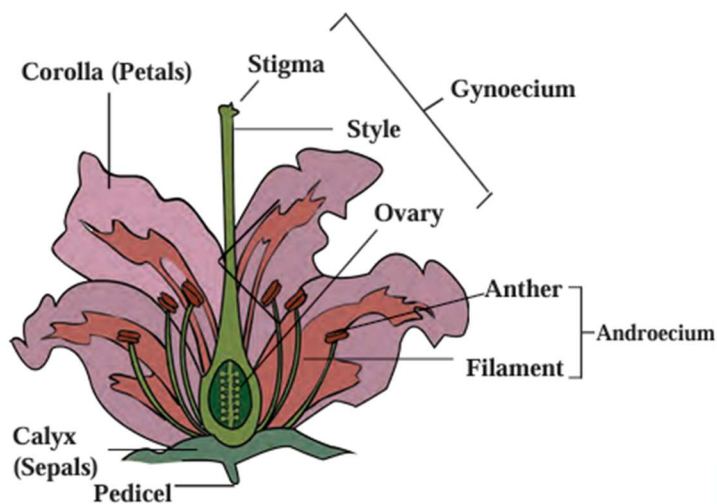
b. Human female reproductive system.

[2024]



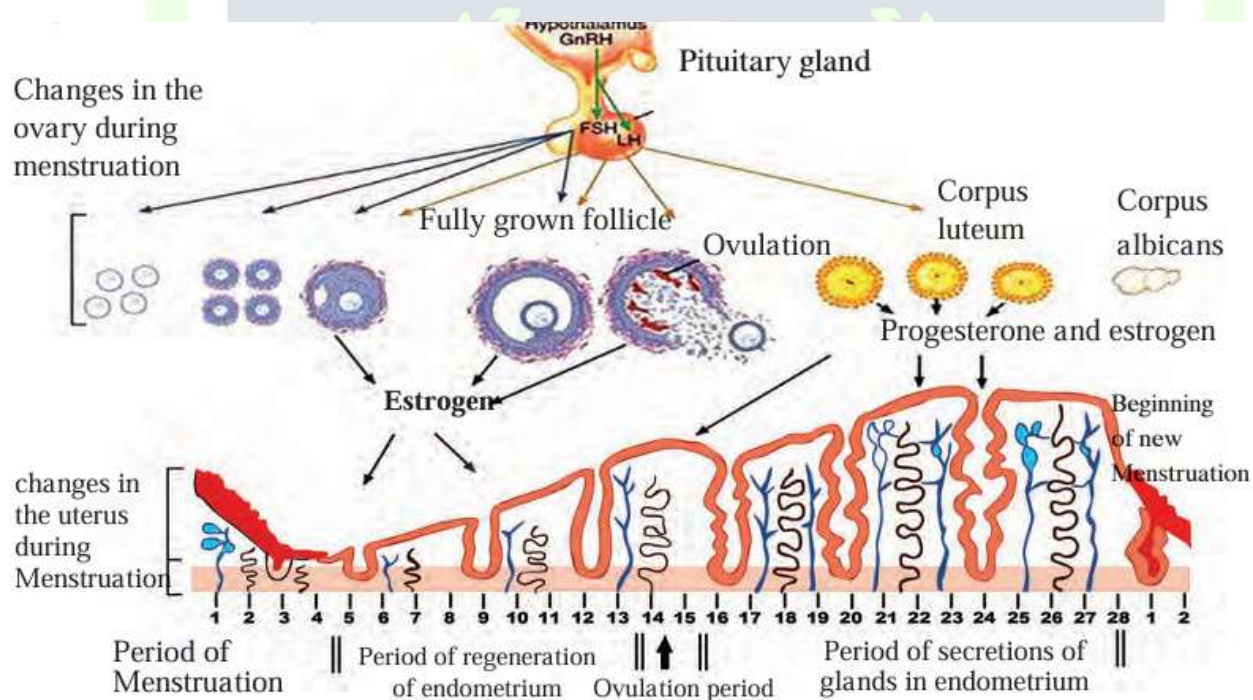
c. Flower with its sexual reproductive organs.

[2022]



TO LEAD THE LEARNERS BEYOND THE STARS

d. Menstrual cycle



3 Marks Questions:

Q. Answer the following questions.

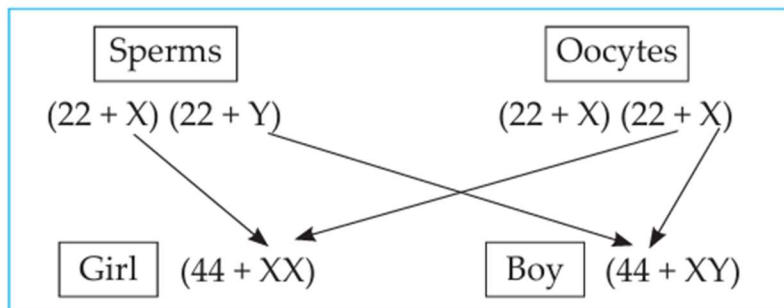
1. Gender of child is determined by the male partner of couple. Explain with reasons whether this statement is true or false.

OR It is not right to consider the mother responsible, for giving birth to a girl child. [2025]

Ans:

1. The man is totally responsible, whether the couple will have a boy or a girl child.

- During zygote formation, man contributes either X or Y chromosome to the next generation, but woman contributes only X chromosome to the next generation.
- At the time of fertilization, if X- chromosomes comes from male, the child will be a girl and if Y- chromosome comes then the child will be a boy.
- Therefore, it is true that gender of child is determined by the male partner of couple.



2. Complete the following chart.

OR Distinguish between Sexual and Asexual reproduction.

[2022, 2024, 2025]

Asexual reproduction	Sexual reproduction
1. Reproduction that occurs with the help of somatic cells is called as asexual reproduction.	1. Reproduction that occurs with the help of male and female gametes is called sexual reproduction.
2. Only one parent is necessary for asexual reproduction.	2. Male and female parent are necessary for sexual reproduction.
3. This reproduction occurs with the help of mitosis only.	3. This reproduction occurs with the help of meiosis and mitosis.
4. New individual formed by this method is genetically identical to the parent.	4. New individual formed by this method is genetically different from parents.
5. Asexual reproduction occurs in different individuals by various methods like binary fission, multiple fission, budding, fragmentation, regeneration, vegetative propagation, spore production, etc.	5. Sexual reproduction occurs through the formation and fusion of male and female gametes.

3. Complete the paragraph with the help of words given in the bracket.

(Luteinizing hormone, endometrium of uterus, follicle stimulating hormone, estrogen, progesterone, corpus luteum)

Growth of follicles present in the ovary occurs under the effect of **follicle stimulating hormone**. This follicle secretes estrogen. **Endometrium of uterus** grows / regenerates under the effect of estrogen. Under the effect of **luteinizing hormone**, fully grown-up follicle bursts, ovulation occurs and **corpus luteum** is formed from remaining part of follicle. It secretes **estrogen** and **progesterone**. Under the effect of these hormones, glands of **endometrium of uterus** are activated and it becomes ready for implantation.

4. Which precautions will you follow to maintain the reproductive health?

[2022]

Ans.

1. While one takes good care of his/her physical health, reproductive health is usually ignored.
2. During the menstrual cycle, a lot of bleeding occurs.
3. Due to this, private organs (genitals) need to be maintained clean from time to time, otherwise problems may arise.
4. Men also need to maintain the cleanliness of their genitals to avoid problems.
5. Proper hygiene must be observed to prevent sexual diseases caused by bacterial infections, such as syphilis and gonorrhoea.
6. After childbirth, in order to let the uterus recover to normal health, the woman must take enough rest and avoid carrying heavy objects.

5 Marks Questions:

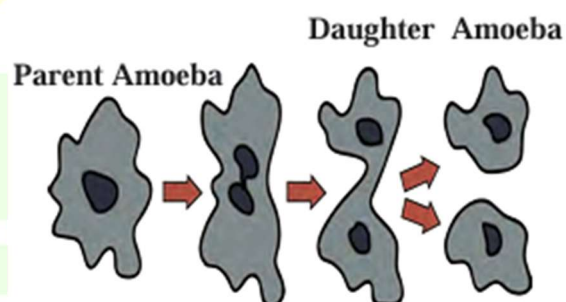
1. Explain with examples types of asexual reproduction in unicellular organism.

[2025]

Ans: Formation of new organisms without involvement of gametes is called asexual reproduction.

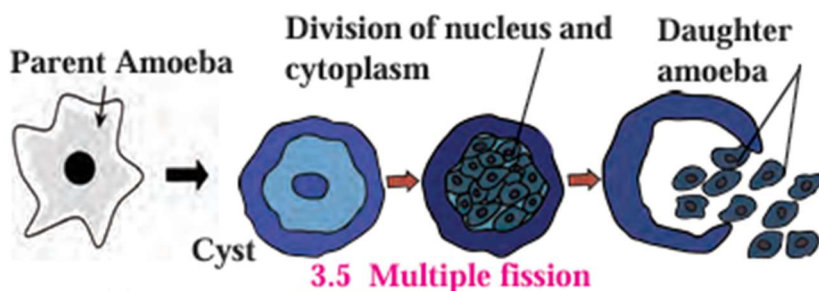
1] Binary Fission:

1. In this process, the parent cell divides to form two similar daughter cells.
2. Binary fission occurs either by mitosis or amitosis



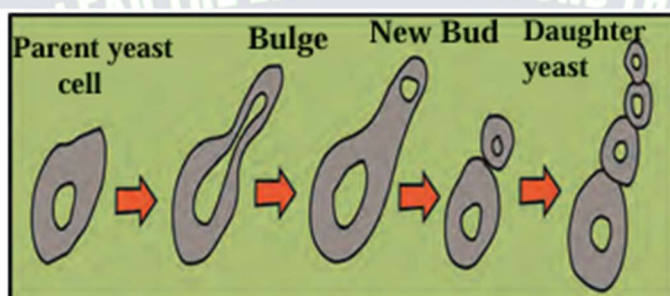
2] Multiple Fission

1. Amoeba becomes rounded and forms protective covering around plasma membrane. Such encysted Amoeba or any other protist is called as 'Cyst'.
2. Many nuclei are formed by repeated nuclear divisions in the cyst. It is followed by cytoplasmic division and thus, many amoebulae are formed.
3. They remain encysted till there are adverse conditions. Cyst breaks open on arrival of favourable conditions and many amoebulae are released.



3] Budding:

1. A small bulge appears on the surface of yeast parent cell. This bulge is actually a bud. One of the two daughter nuclei enters this bud.
2. After sufficient growth, bud separates from the parent cell and starts to live independently as a daughter yeast cell.



II] Extra - Previous Years' Questions: (PYQs)

★ 1 Mark Objectives: ★

1. MCQS:

1. Regeneration occurs in _____. [2024]

- (A) Amoeba (B) paramoecium (C) euglena (D) planaria

2. Type of sexual reproduction is _____. [2023]

- (A) fragmentation (B) regeneration (C) gamete formation (D) budding

2. Odd one out:

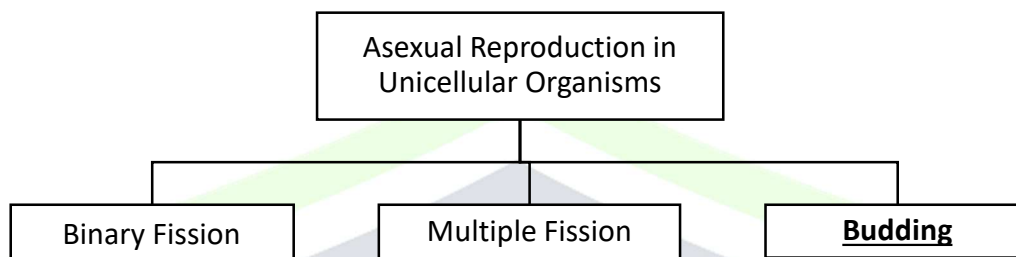
Fragmentation, Regeneration, Budding, **Fertilization** [2025]

3. Match the pair: [2024]

Column A	Column B
Male	a. $44 + XX$ b. $44 + XY$

4. Fill in the box with proper words.

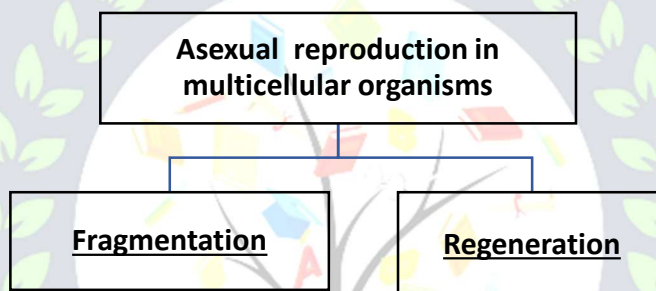
[2026]



2 Marks Questions:

1. Fill in the box with proper word:

[2025]



2. Give scientific reason:

1. Flower is the unit of sexual reproduction.

[2023]

Ans:

- In flower, the androecium and gynoecium are the male and female parts, respectively.
- The carpel is the female reproductive organ, containing the ovary with ovules that form the embryo sac.
- The stamen is the male reproductive part, producing male gametes.
- Pollination transfers pollen from male to female parts, primarily to the stigma. Therefore, flower is the unit of sexual reproduction.

5 Marks Questions:

1.

[2024]

- Draw well labelled diagram of female reproductive system. Refer pg.2 Q.2 b
- Which type of hormone are secreted in female reproductive system. Refer pg.1 Q.1 b
- When does menopause period stop.

Ans: Menopause period stops at the age of about 45-50 years.

d. How many ovaries are there in female foetus at the time of birth?

Ans: Two ovaries are there in female foetus at the time of birth.

2. Answer the following questions based on human male reproductive system.

[2024]

a. In which organs sperms are formed?

Ans: Sperms are formed in testes.

b. What is the length of a sperm?

Ans: The length of a sperm is about 60 micrometres.

c. What is the work of vas deferens?

Ans: Vas deferens transfers sperm cells from the epididymis to urethra before ejaculation.

d. By which type of cell division, sperms are formed?

Ans: Sperms are formed by meiosis type of cell division.

e. Name any two unpaired organs of male reproductive organs?

Ans: Prostate glands and penis are unpaired organs in male reproductive system.

3. Answer the following questions based on human female reproductive system.

[2023]

a. How many immature oocytes are present at the time of birth in the ovary of female foetus?

Ans: 2 – 4 million oocytes are present at the time of birth in the ovary of female foetus.

b. What is ovulation?

Ans: Bursting of mature ovarian follicle under the influence of luteinizing hormone to release the oocyte is called ovulation.

c. Draw scientifically correct diagram of female reproductive system.

Ans: refer page no. 2, Q. 2. B

d. Label the following parts: **Ans:** refer page no. 2, Q. 2. B

- i. Ovary
- ii. Uterus

4. Answer the following questions by observing the diagram of menstrual cycle.

[2023]

For diagram refer page. no. 3 Q. 2. D

a. What is the period of menstruation?

Ans: 1 to 5 days is the period of menstruation.

b. On which day does ovulation occur during menstrual cycle?

Ans: Ovulation occurs between 14th to 15th day.

c. In menstrual cycle which reproductive organs undergo changes?

Ans: The ovary and the uterus change during the menstrual cycle.

d. Which period is said to be period of regeneration of endometrium?

Ans: In the menstrual cycle, day 5 to 14 are the period of regeneration of endometrium.

e. Which period is said to be period of secretion of glands in endometrium?

Ans: The period of secretions of glands in endometrium is 15 to 28 days.

5. Draw a neat labelled diagram of female reproductive system and give the function of [2022]
refer page no. 2, Q. 2. B for diagram.

a. Oviduct

Oviduct or fallopian tubes is the place through which eggs travel from ovary, meet sperms and fertilization takes place.

b. Ovary

The ovary produces female gamete or egg and the female hormone estrogen.

c. Uterus

The uterus accommodates the growing foetus till it completes growth.

6. [2022]

a. Sketch and label the essential and accessory whorls of flower?

Ans: *refer page no. 3, Q. 2. C*

b. What is pollination?

Ans: Transfer of pollen grains from male anther to female stigma in flowers is called pollination.

c. Give any two examples of agents of pollination.

Ans: The agents of pollination are wind, water, insects and animals.

7. Observe the diagram and answer the following questions. [2026]

Refer pg.2 Q.2 A

1. Write the name of the reproductive system in the above diagram.

Ans: Human male reproductive system.

2. Name any two glands in the above diagram.

Ans: Prostate gland and Cowper's gland.

3. Mention any one duct in the above system.

Ans: Ejaculatory duct.

4. What is the length of a sperm?

Ans: The length of a sperm is 60 micrometres.

5. Which hormone is secreted by the testes.

Ans: Testosterone

